

Isolation Forest: A Machine Learning Approach to Fraud Detection

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Abstract—Isolation Forest (iForest) has emerged as a foundational technique for unsupervised anomaly detection. iForest is particularly appealing for its efficiency, simplicity and performance with unlabeled data. Since its inception, iForest has evolved into a broad ecosystem of variants that span classical, streaming, high-dimensional, hybrid, deep-learning, and explainable models. This review examines these subcategories with an emphasis on their application to real-world fraud detection tasks. Although iForest is often regarded as an efficient lightweight anomaly detection method, we argue that its utility in fraud scenarios is limited by insufficient transparency, inadequate support for streaming data, and a lack of investigation into the subtleties of fraud. By analyzing a diverse body of recent literature and studies, we identify gaps in the field and propose key directions to align Isolation Forest research with the complex demands of real-world fraud detection.

Index Terms—isolation forest, anomaly detection, financial fraud detection, streaming data, machine learning

I. INTRODUCTION

Isolation Forest (iForest) is a machine learning method created with the intention of efficiently detecting anomalies in unlabeled data. Since its introduction in 2008 by Liu et al. [1], iForest has been applied in a wide range of domains, including fraud detection, quality control, sensor monitoring, and medical diagnostics. Unlike other forms of anomaly detection that often rely on discriminative methods that are rule-based, iForest operates on the principle that anomalous data points are easier to isolate and therefore tend to appear closer to the root of the binary isolation trees. Due to the nature of this isolation mechanism, iForest does not attempt to classify individual data points directly; instead, it measures how easily each point can be separated from the rest of the dataset using random feature splits. This approach allows it to perform well in high-dimensional, unlabeled, and imbalanced settings, making it particularly appealing in domains where labeled anomalies are scarce [2].

Previous work has systematically reviewed iForest literature [3]; however, there is currently no unified review that critically

assesses how iForest variants address the practical demands of fraud detection, such as streaming capability, categorical feature handling, and interpretability for audit situations. This review focuses on methods that address the demands of fraud detection, such as model robustness [4]–[8], the ability to handle streaming data [9], and the ability to explain the output of the model [10], [11]. Despite the depth of literature in these domains, there is no apparent solution that encompasses these requirements; we believe that the fragmentation of iForest research limits our understanding of iForest’s true capability in fraud detection and highlights the need for a comparative analysis.

To address this gap, we offer a structured analysis of iForest research tailored to fraud detection, organized as follows.

- In Section II, we introduce the basics of fraud detection, establish industry requirements, and dive into the domain of financial anomaly detection.
- In Section III, we discuss the core operation of iForest as an anomaly detection mechanism.
- In Section IV, we investigate model trade-offs while categorizing 24 methods from recent research into six areas: core algorithms, streaming variants, high-dimensional adaptations, hybrid approaches, deep-learning integrations, and explainable approaches.
- In Section V, we evaluate how well these iForest variants and studies align with the domain requirements established in Section II.
- In Section VI, we discuss the complexities of the open technical challenges, observe where the gap lies, and propose a possible future research domain.

By placing increased scrutiny on the capabilities of iForest in fraud detection, we advocate for the realignment of iForest research towards techniques that are interpretable and capable of handling streaming data, while maintaining the lightweight framework.

II. BACKGROUND

A. Fraud Detection and Industry Requirements

Before diving into the vast literature of anomaly detection, we define the concepts of fraud and fraud detection. Fraud can be defined plainly by the Cambridge Dictionary as "the crime of getting money by deceiving people." According to Bolton and Hand [12], fraud detection refers to the process of monitoring and analyzing transactions to identify those that are likely to be fraudulent. Fraud detection, rather than trying to prevent fraud, focuses on recognizing suspicious activities after or during the occurrence, allowing for timely investigation and mitigation.

Fraud detection is of critical importance not only for financial institutions but also for e-commerce, insurance, and cybersecurity domains, where fraud causes substantial economic losses and damages reputations. Globally, fraud is estimated to cost companies and individuals over \$5 trillion annually [13]. In the United States alone, data from 2018-2022 shows the federal government loses between \$233 to \$521 billion annually to fraud [14].

Bolton and Hand [12] further emphasize that fraud detection is inherently challenging due to the dynamic and adaptive nature of fraudulent behavior, the extreme imbalance between legitimate and fraudulent cases, and the requirement for real-time decision making. They identify several factors essential for effective fraud detection, including:

- **Accurate modeling of normal behavior**, providing a reliable baseline for anomaly detection;
- **Adaptability to evolving fraud tactics**, as fraudsters continuously adapt their approaches to the new financial and technical landscape;
- **Scalability**, ensuring detection systems can handle massive transaction volumes efficiently; and
- **Timeliness**, enabling rapid responses to suspicious activities.

Beyond just the technical requirements of fraud detection, in recent years, the fraud detection domain has shifted towards explainable artificial intelligence (XAI) in fraud detection, noting that interpretability is crucial not only to meet regulatory standards but also to foster trust and adoption within financial institutions [15]. Financial regulators around the world now expect AI systems to be transparent, especially in high-stakes domains like money laundering detection, credit scoring, and fraud detection. This expectation of explainability is reflected in the EU's GDPR Article 22, Recital 71 which gives the user the right "to obtain an explanation of the decision reached after such assessment and to challenge the decision" [16] when an automated decision mechanism is employed. Regulators in the US, while not directly pointing to the complete auditability of AI models as a requirement, require that all regulated entities be able to explain their use of AI models [17] when explaining their decisions to experts and regulators.

B. Anomaly Detection in Fraud Detection

The technical challenges outlined by Bolton and Hand demonstrate that traditional fraud detection methods, such as rule-based models and supervised learning models, often fall short in dynamic fraud environments. Rule-based approaches depend on predefined patterns, making them less effective against novel fraud schemes, while advanced supervised models are constrained by the availability and totality of labeled fraud data. As fraud tactics evolve rapidly, detection systems must not only meet accuracy and scalability requirements, but also adapt to emerging threats. This need for adaptability and the ability to uncover previously unseen fraudulent behavior position **anomaly detection** as a critical methodology in modern fraud detection frameworks.

Anomaly detection is a field of research that investigates methods to identify if an observation deviates from what is expected of that observation. Following the formal definition by Ruff et al. [18] an anomaly can be defined as follows:

Let $X \subseteq \mathbb{R}^D$ denote the data space associated with a given observation. The concept of normality is characterized by a probability distribution P^+ on X that is the ground-truth law of normal behavior. An anomaly is then defined as any observation $x \in X$ that lies in a low probability region under P^+ . Assuming that P^+ has a probability density function (pdf) $p^+(x)$, the set of anomalies can be defined as:

$$A = \{x \in X | p^+(x) \leq \tau\}, \tau \geq 0 \quad (1)$$

where τ is a threshold chosen such that the probability of A under P^+ is sufficiently small.

This probabilistic formulation highlights that anomaly detection methods do not rely on explicit rules for classifying data points as anomalous; instead, they focus on modeling a distribution of normal behavior and flagging deviations from this norm. Such an approach is well-fit for domains where anomalous patterns are rare, diverse, and evolving, such as fraud detection.

In the context of fraud detection, anomaly detection is used to identify transactions that deviate from the expected distribution of legitimate transactions. Unlike traditional rule-based or supervised approaches that rely on predefined fraud indicators or labeled examples, anomaly detection models learn the distribution of normal transactions, creating a baseline, and flag those falling into low-probability regions of this learned distribution. Based on this principle, Bolton and Hand [19] propose calculating suspicion scores for each account in a database, with the highest scoring accounts prioritized for further investigation.

Since the work of Bolton and Hand in 2001, the landscape of anomaly detection in the financial domain has evolved significantly, incorporating more sophisticated techniques. In a comprehensive survey, Ahmed et al. [20] examined anomaly detection methods applied to financial data with a focus on clustering-based models. The key conclusion of this survey is that anomaly detection methods applied to financial data remain limited because the majority of these studies rely

on synthetic data, leading to possible gaps in real-world deployment. Furthermore, a universal technique in the domain of fraud detection is yet to be found, due to the evolving context and scarcity of labeled data.

These challenges have motivated the development of alternative anomaly detection algorithms, beyond clustering, designed to operate effectively with limited labeled data and dynamic fraud behavior. Among these, **iForest** is a prominent method due to its ability to isolate anomalies efficiently without relying on extensive labeled datasets.

III. THE PRINCIPLE OF OPERATION OF IFOREST

The isolation forest, introduced by Liu et al. in 2008 [1], represents a change in anomaly detection by approaching the problem through the lens of isolation rather than constructing a profile of normal instances. Traditional anomaly detection methods, including clustering and statistical techniques, assume that anomalies are points far from the center of known clusters. In contrast, Isolation Forest explicitly leverages the observation that anomalies are both *few* and *different*; thus, they can be isolated from the majority of data points more easily than normal instances.

Architecturally, iForest constructs an ensemble of binary trees, known as isolation trees (iTrees), by recursively partitioning the data through randomly selected features and split values. The key insight is that anomalies, being rare and distinct, tend to have shorter path lengths in these trees because they can be separated with fewer splits. The anomaly score is then derived from the average path length of a point, with shorter paths indicating a higher probability of anomaly. This tree-based mechanism eliminates the need for calculating, sometimes complex, distance measures or density estimates.

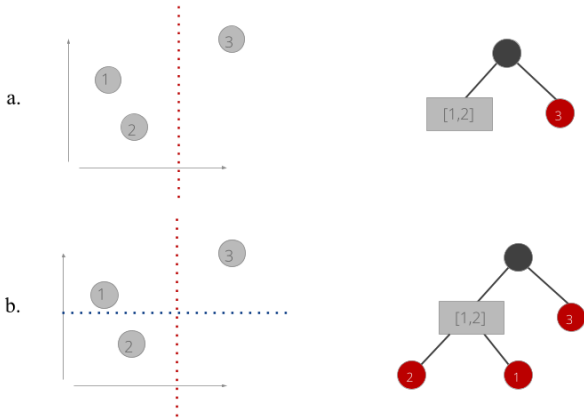


Fig. 1. Construction of an iTree

Figure 1 shows the construction of an iTree for a dataset that consists of three data points. When iForest is deployed, an ensemble of iTrees is constructed with randomly selected subsamples, split direction, and split location. During evaluation, an anomaly score is calculated by finding the path that the test point would take down each of the iTrees, with a shallower path resulting in a greater addition to the anomaly score.

Liu et al. demonstrated that iForest not only achieves competitive or superior detection performance, averaging 0.87 Receiver Operator Characteristic (ROC) Area Under the Curve (AUC) across 12 benchmarks, compared to clustering-based and density-based approaches, such as ORCA, Local Outlier Factor (LOF) and Random Forest (RF), but also offers substantial gains in efficiency. iForest performed efficiently on all benchmarks because it scales linearly with the size of the data set. This characteristic allows it to perform particularly well in high-volume transaction environments, such as those typically seen in financial fraud detection. An interesting occurrence is that the Pima and Ionosphere datasets consisted of the highest prevalence of anomalous data at 35% and 36% respectively, and these are the only two benchmarks where iForest did not perform the best. This shows that iForest is a strong candidate for fraud detection because fraud occurs at relatively low rates, just like in the datasets where iForest excelled. Furthermore, because iForest does not rely on labeled data, it addresses one of the key challenges highlighted by Ahmed et al. [20] – the scarcity of labeled fraud datasets. Its unsupervised nature, combined with computational efficiency, adaptability, and robustness in high-dimensional spaces, explains why iForest has become a widely adopted tool in modern fraud detection pipelines.

Although iForest is a strong candidate for fraud detection because of its strength in identifying anomalous data in scarce data sets, low computational complexity, and performance in an adaptive environment, it still falls short with streaming data, high-dimensional data, and in terms of explainability. In the years since the introduction of iForest in 2008, many variants have been developed that improve efficiency, model performance, and explainability. These developments have created a vast landscape of iForest research.

IV. THE EVOLVING ISOLATION FOREST ECOSYSTEM: TECHNICAL ADVANCES & PRACTICAL GAPS

The introduction of iForest by Liu et al. [1] not only transformed anomaly detection by focusing on isolation, but also catalyzed a wave of research aimed at extending and refining its capabilities. Over the years, this research has diverged into several subdomains to address the fundamental challenges of real-world data, such as imbalanced data, adaptive learning, robustness, and explainability. All of these challenges are vital for the field of financial fraud detection.

The following sections critically examine the algorithms that define this evolution, beginning with the original iForest and its most widely adopted extension, the Extended Isolation Forest (EIF). Through this examination, we highlight how incremental algorithmic innovations have shaped current practices and where practical gaps remain. Fig. 2 shows the taxonomy of iForest research and breaks it down by the challenges each subdomain directly confronts.

A. Core Algorithms

iForest, introduced in 2008 by Liu et al. [1], was a defining moment in anomaly detection research. The key break-

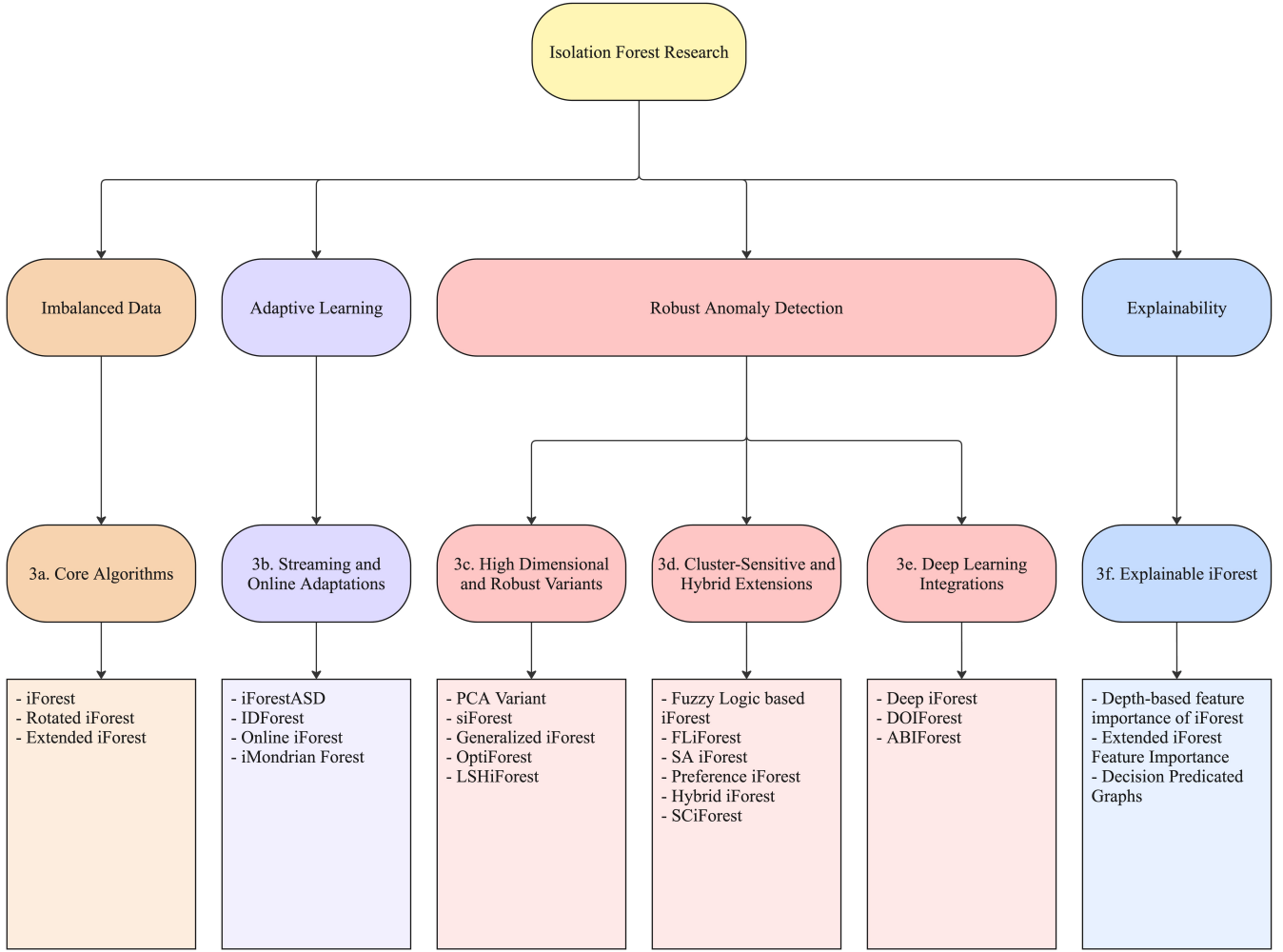


Fig. 2. Taxonomy of iForest Research

through lies in using random partitioning to isolate anomalies rather than explicitly modeling normal data distributions via clustering or other statistical methods. This approach makes iForest computationally efficient, scaling linearly with dataset size, while remaining robust and effective with unbalanced unlabeled data. Beyond its initial breakthrough, iForest has remained a cornerstone in anomaly detection research and practice for more than a decade. Its unsupervised framework requires minimal parameter tuning for widespread deployment. Furthermore, its ensemble structure minimizes overfitting and makes it highly adaptable to dynamic environments, as we would see in financial fraud detection, where patterns of anomalies evolve rapidly. These characteristics of iForest have ensured its adoption as a baseline in both academic studies and industry applications. In domains where anomalies exhibit complex structures, iForest often serves as a strong benchmark due to its efficiency and competitive detection performance. These enduring strengths show why iForest continues to be

widely cited and implemented, despite the emergence of improved variants. Despite these strengths, iForest’s reliance on axis-aligned splits introduces a structural bias: anomalies that are not easily separable along individual feature axes may not receive consistent anomaly scores.

Rotated Isolation Forest (RIForest) was first proposed by Hariri et al. [21] as an early method to address the linear split structural bias. Instead of restricting partitions to axis-parallel cuts, RIForest applies random rotations to the feature space before building each isolation tree. These rotations increase the likelihood that anomalies distributed along arbitrary orientations are effectively isolated, improving detection in cases where fraudulent or anomalous behavior exploits feature correlations. Although this enhancement reduces the structural bias of iForest, it also introduces drawbacks: repeated transformations increase computational overhead, and instability may occur on asymmetric datasets, leading to inconsistent anomaly scores. Recent work by Monemizadeh et al. [22] has sought to

stabilize RIForest by addressing issues such as ‘ghost clusters’ in high-dimensional settings, but computational efficiency remains a concern. Moreover, evidence suggests RIForest can overpredict anomalies, raising the risk of false positives in fraud detection contexts where legitimate transactions vastly outnumber fraudulent ones.

Extended Isolation Forest (EIF) [21] is a later method developed by Hariri et al. that generalizes the partitioning strategy by introducing splits along randomly sampled hyperplanes rather than relying on rotations of the input space. EIF eliminates the bias toward axis-aligned structures while avoiding the instability associated with repeated rotations. By allowing cuts at arbitrary orientations, EIF produces smoother, more reliable anomaly scores and consistently outperforms iForest on datasets with nonlinear or complex feature interactions. Comparative studies demonstrate that while EIF performs similarly to iForest in well-separated, low-dimensional settings, it excels with real-world datasets where anomalies are embedded in heterogeneous or correlated spaces – shown in Table I. Importantly, EIF achieves these gains without sacrificing the scalability and efficiency of iForest.

TABLE I
AUC VALUES FOR BOTH ROC AND PRC IForest VS EIF IN
BENCHMARK DATASETS ADAPTED FROM [21]

Benchmark	AUC ROC		AUC PRC	
	<i>iForest</i>	<i>EIF</i>	<i>iForest</i>	<i>EIF</i>
Cardio	0.888	0.915	0.466	0.483
ForestCover	0.809	0.924	0.430	0.504
Ionosphere	0.85	0.913	0.877	0.893
Mammography	0.859	0.862	0.4198	0.4271
Satellite	0.714	0.778	0.783	0.808

The transition from iForest to RIForest and ultimately to EIF illustrates how incremental algorithmic refinements can yield substantial improvements in robustness and detection power. Table II summarizes the trade-offs: iForest provides scalability and simplicity at the cost of structural bias; RIForest reduces that bias but suffers from computational overhead and instability; and EIF delivers the most balanced approach, achieving low bias and high stability while retaining competitive efficiency.

TABLE II
COMPARISON OF IForest VARIANTS

Algorithm	Scal.	Bias	Stable	Complexity	Non-linear Perf.
iForest	High	High	High	Low	Low
Rot. iForest	Med.	Low	Low	High	Med.
EIF	High	Lowest	High	Med.	High

In summary, while iForest remains a strong and widely used baseline, its axis-aligned bias motivated refinements that improved detection in complex domains. RIForest offered the first step toward greater geometric flexibility, though at the cost of efficiency and stability. EIF resolved these issues by introducing hyperplane-based partitions, ultimately establishing itself as the most effective and general-purpose variant. These

advances set the stage for subsequent research that integrates adaptability, feature weighting, and interpretability – directions explored in the following subsections.

B. Streaming and Online Adaptations

Due to the evolving landscape of fraud and the advanced tactics groups use to repeatedly disguise their true identities and modify their *modus operandi* [23], it is important for fraud detection mechanisms to update online as streaming data arrives. To confront the issue of streaming data, online variants of iForest have been introduced to the research landscape, including: iForestASD [9], IDForest [24], Online iForest [25] and iMondrian Forest [26].

iForest, while not originally developed for streaming data, performs well when tested in an adaptive environment. Kareem et al. [2] demonstrated that the iForest framework, despite its traditionally offline nature, exhibits strong performance in streaming environments by leveraging its efficient isolation-based partitioning. This study highlights the performance of iForest in a real-time context and lays the foundation for building iForest variants that incorporate new data incrementally to adapt to evolving distributions, a trait that is vital for model performance in fraud detection.

iForestASD [9] introduces an adaptive sliding window subsampling mechanism that incrementally retrains the iTrees as new data arrive. iForestASD functions by implementing a sliding window that stores N data points; the model then calculates anomaly scores on the N data points and outputs the results. For the online updating component there is a threshold u that establishes when the model should retrain on the currently held data because a concept drift is apparent. If $n > u$, n being the number of anomalous data points in the N samples, then the model retrains. In practice, this threshold-driven retraining allows iForestASD to react quickly to sudden changes in data distribution, maintaining competitive detection accuracy while avoiding excessive computation. However, in fraud detection scenarios where malicious behavior can change in subtle, long-term ways rather than sharp bursts, this model may fail to capture slow shifts in fraudulent activity until they cross the anomaly threshold, potentially leaving extended detection blind spots.

IDForest [24] is built upon LSHiForest (covered in Section 4c.) and is intended for edge device deployment with limited memory, thus it incrementally inserts and deletes data to support online adaptation. IDForest relies on a fixed maximum tree size and implements the following algorithm for online updating. When point p is introduced into the dataset the distribution will shift, and p is added to the tree in one of two ways, 1. there exists a leaf with the same index and the data point joins that leaf or 2. a new leaf is created. When the maximum tree size has been reached, the deletion function takes place leading to the removal of a leaf in a FIFO manner and rebalancing the tree. Through this insertion and deletion process the IDForest consistently has the most relevant data points present in its iTrees, making it adaptive in nature. The lightweight update mechanism makes IDForest efficient for

real-time, resource-constrained environments while retaining strong detection rates in fast-changing data streams. Yet, for fraud detection, the FIFO deletion strategy risks removing rare but informative fraudulent patterns before they can be fully leveraged, particularly in class-imbalanced settings where legitimate transactions vastly outnumber fraudulent ones.

Online iForest [25] builds on the principles of iForestASD by also implementing a sliding window of data points for training the model. The key differences lies in the retraining with all new data points rather than just at a specific anomaly threshold. Additionally, Online iForest increases the bin height whenever the limit is met, it does this so regions with more data points have a higher resolution and can capture more complex trends. Like the previous two algorithms, Online iForest also has a deletion algorithm triggered by bin height. The deletion procedure in this algorithm reduces model complexity by merging nodes and their associated bins when older data points are processed, effectively lowering resolution in that region. As the bin height of a split node drops below a threshold, its two child nodes merge. This continuous retraining approach makes Online iForest particularly sensitive to evolving trends and localized anomalies, giving it strong adaptability in dynamic environments. However, in fraud detection, frequent retraining on every new transaction risks overfitting to short-term fluctuations in legitimate behavior, the same problem seen in iForestASD, reducing its ability to distinguish slow-developing fraud strategies that span multiple windows.

iMondrian Forest [26] introduces an online learning mechanism based on the Mondrian process. Unlike previous algorithms that rely on windowed retraining or deletion strategies, iMondrian Forest dynamically adjusts its tree structure by extending nodes whenever new data points arrive. When a new point falls outside the current bounding region of a node, the algorithm samples an exponential variable to determine whether to create a new parent split above the existing node. If the condition is met, a new split dimension and split value are sampled and the new point is inserted as either a left or right child. If the point lies within the bounds, the algorithm simply descends the tree recursively without altering existing splits. This design enables iMondrian Forest to expand partitions adaptively and maintain high-fidelity representations of streaming data without full retraining or fixed window sizes. Its ability to incrementally refine partitions without discarding older data makes it resistant to concept drift in many domains. But in fraud detection, where adversaries may deliberately craft transactions that remain within known distributions to evade detection, the lack of regular repartitioning or pruning could allow persistent fraudulent behavior to remain hidden.

The comparison in Table III illustrates the state of streaming iForest variants. iForestASD and Online iForest effectively manage the memory the model utilizes by employing a sliding window that the model is trained on. In contrast, IDForest employs a method of insertion and deletion when creating the trees to ensure the model is friendly to edge devices. Lastly, iMondrian Forest dynamically refines tree partitions without discarding historical data, supporting adaptive learning

TABLE III
FEATURE COMPARISON OF ONLINE iFOREST VARIANTS

Feature	iForestASD	IDForest	Online iForest	iMondrian
Sliding Window	✓		✓	
Deletion		✓		✓
Merging		✓	✓	
Adaptive Resolution			✓	
Probabilistic Update				✓
Retraining	✓		✓	
Incremental Insertion		✓		✓
Handles Concept Drift	✓		✓	✓
Low Memory Usage	✓			
Edge Friendly		✓		

over time. Despite these advancements, each approach presents trade-offs in balancing memory efficiency, update frequency, and robustness to concept drift, all of which must be carefully considered when deployed in fraud detection systems.

C. High Dimensional and Robust Variants

An issue when working with real-world fraud datasets is the high dimensionality of the data. To account for high-dimensional data, researchers have developed a multitude of robust variants of iForest. Among these variants of iForest are PCA Variant iForest [6], siForest [27], Generalized iForest [28], LSHiForest [29], and OptiForest [30].

PCA Variant iForest [6] is a method of using Principle Component Analysis (PCA) pre-processing of data paired with iForest. PCA Variant iForest functions by performing PCA on the high-dimensional input data prior to training and evaluation. The results of the study showed improved accuracy over iForest and claimed increased efficiency. The efficiency claims of this study remain unverified, and if the run time of the PCA exceeds the run time of the iForest algorithm, this method would reduce the real-time benefits of iForest. Additionally, PCA reduces high-dimensional features into orthogonal components capturing the variance of the original data, greatly reducing the interpretability of the results.

Set-Partitioned Isolation Forest (siForest) [27] is another modification of iForest aimed at analyzing complex, multidimensional datasets. siForest modifies iForest by stopping the splitting behavior once all data points belong to a set. This concept could be applied in the domain of fraud detection by identifying where or with what user an anomaly occurred and then conducting a deeper investigation into the group the model identified. This would allow for rapid detection of anomalous sets prior to running a more computationally complex investigation into the anomaly.

Generalized iForest (GIF) [28] aims to generalize the iForest framework beyond the path length metric for anomaly scoring. Rather, GIF estimates the probability distribution of the spatial data and detects anomalies by finding regions with small mixture weights, thus indicating sparsity. It does so by partitioning the feature space into non-overlapping regions via iTrees, computing the empirical density in each region, and ranking regions by their mixture weights, where low-weight

regions are flagged as potential outliers. This formulation ties isolation-based detection to density estimation, enabling GIF to target rare subpopulations in the data that traditional path length metrics may overlook. GIF yields results in line with EIF and iForest and sometimes exceeds their performance; furthermore, GIF claims comparable run time to these methods. Its interpretable density-based scoring, competitive accuracy, and run time make it suitable for explainable and real-time fraud detection scenarios. However, GIF’s reliance on accurate density estimation within high-dimensional spaces can degrade performance when data is sparse, an occurrence common in fraud datasets.

Locality Sensitivity Hashing iForest (LSHiForest) [29] is designed to enhance the detection of anomalies in high-dimensional data by incorporating a locality-sensitive hashing (LSH) scheme into the traditional iForest framework. By projecting data into a lower-dimensional hash space using LSH, LSHiForest groups similar instances together, reducing the dimensionality of data and improving the efficiency of partitioning. These hashed groups are then processed through iTrees, enabling faster separation of anomalous points while preserving the core isolation concept. Due to the nature of hashing, the *iTrees* can split in a multisplitting manner rather than a binary manner, greatly speeding up the tree traversal. Empirical results show that LSHiForest achieves superior accuracy compared to standard iForest in most cases on high-dimensional datasets, with notable gains in speed. Its hashing-based preprocessing greatly speeds up run time, making it perfect for a real-time application. However, its reliance on hash function parameters introduces sensitivity to tuning, which can impact performance in datasets with complex, overlapping fraud patterns, as well as issues of interpretability.

OptiForest [30] builds upon the hashing and multi-splitting concepts of LSHiForest, extending them by introducing a method for constructing iTrees with an optimal branching factor. Instead of relying solely on binary splits, OptiForest adaptively determines the best number of branches at each node to maximize separation between normal and anomalous instances. This is achieved through an optimization process that evaluates split quality in the hashed feature space, creating more balanced partitions and reducing tree depth. Experimental results indicate that OptiForest can outperform LSHiForest and iForest in both detection accuracy and run time efficiency, especially in high-dimensional settings. While its efficiency and scalability make it appealing for time-sensitive anomaly detection tasks, the additional optimization layer and hashing step reduce transparency in the decision process and increase the complexity of parameter tuning.

D. Cluster Sensitive and Hybrid Extensions

Building upon advances in high-dimensional variants of iForest, researchers have developed cluster-sensitive and hybrid extensions. Cluster-sensitive approaches integrate clustering mechanisms to better capture local data structures and relationships, improving detection of anomalies that deviate

TABLE IV
FEATURE COMPARISON OF HIGH-DIMENSIONAL iFOREST VARIANTS

Feature	PCA-iForest	siForest	GIF	LSHiForest	OptiForest
Dimensionality Reduction	✓				✓
Set Partitioning / Subspace Split		✓		✓	✓
Generalized Splitting Criterion			✓		
Hash-Based Projection				✓	✓
Optimization for Split Selection					✓
Computational Efficiency	Moderate	High	Moderate	High	Moderate
Robustness to Irrelevant Features	High	High	High	High	High

from cohesive groups. Hybrid methods combine iForest with other machine learning techniques to enhance detection accuracy, scalability, or interpretability. Collectively, these extensions represent a move toward models that not only improve performance but also provide deeper insights into the nature and location of anomalies. Among the models that make this leap are Fuzzy Set-Based iForest [31] [32], FLiForest [5], SA iForest [4], Preference iForest [33], Hybrid iForest [34], and SCiForest [35].

Fuzzy logic-based iForest adaptations such as **Fuzzy Set-Based iForest** [31] and **FForest** [32] combine fuzzy logic with iForest to detect more complex patterns than possible with binary splitting. These methods apply triangular membership functions to quantify how well a point fits within each cluster. During tree construction, each node represents a cluster, and successive splits produce smaller subclusters, making the position of a data point more precisely defined as the traversal deepens. This fuzzy approach can improve accuracy and reduce tree depth compared to iForest, while also offering a modest degree of decision interpretability. However, these methods still struggle with highly sparse anomalies; in such cases, anomaly scores may be diluted due to low membership values across clusters.

FLiForest [5] aims to increase iForest performance in high-dimensional data by weighing each feature and finding a local neighborhood anomaly score. Feature weights are derived using a Maximum-Relevance-to-Density-based approach, which prioritizes attributes most correlated with density variations while still allowing weakly correlated features to contribute. During tree construction, features are selected with probability proportional to their assigned weight, reducing the influence of irrelevant dimensions. The anomaly score is improved by averaging each point’s score with those of its nearest neighbors for the detection of local anomalies and to mitigate false positives. The performance benchmarks demonstrated that FLiForest improved the ROC AUC by 8%-30%. This study, like the fuzzy logic studies, also takes a step towards interpretability of decisions.

Simulated Annealing iForest (SAiForest) [4] improves iForest by reducing the stochastic nature of iForest by pruning lower performing trees. It begins by building an initial iForest, then evaluates each tree’s accuracy (via cross-validation) and diversity (via Q-statistics), combining them into a fitness score. Using simulated annealing, the algorithm selectively replaces trees, sometimes accepting less fit candidates early on to balance exploration and exploitation. This iterative process

yields a more accurate, and more diverse iForest. While the performance of this model exceeds iForest and other state-of-the-art methods, the increased number of hyperparameters might make it difficult to tune and redeploy the model in real-time environments.

Preference iForest (PIF) [33] builds on iForest by adding a preference embedding and a preference isolation function. In the preference embedding part of the algorithm, PIF maps the raw data into the preference space by seeing how well it fits in the modeled spaces of genuine data. The isolation function then operates within this transformed space, using distance metrics tailored to preference vectors to better separate anomalies from genuine points. By isolating points in a space where anomalies become sparse and nearly orthogonal to normal data, PIF improves detection accuracy in complex and structured datasets. The work of Leveni et al. investigates optimizations of PIF to ensure computational efficiency. However, PIF still depends heavily on deducing a model family prior distribution $\mathcal{F}$.

Hybrid iForest [34] builds on iForest by addressing a 'blind spot' that arises when anomalies are embedded within concave regions of normal data. To overcome this, Hybrid iForest employs distance metrics in two ways: first, by calculating the centroid of samples in each external node when the maximum depth of an iTree is reached and incorporating the distance from a point to this centroid into its anomaly score; second, by computing centroids of known anomalies and measuring a point's distance from these centroids. This dual approach improves detection when anomalies are surrounded by dense normal data. However, the method introduces additional computational overhead and may lose effectiveness when anomalies are not spatially separable from normal points in the transformed space; furthermore, the second addition is ineffective if anomaly data are unlabeled.

Isolation Forest with Split-selection Criterion (SCiForest) [35] builds an iForest that goes beyond the assumption of distant and sparse anomalies. SCiForest separates clustered anomalies from normal points by introducing a split-selection criterion that favors partitions that maximize the gap between potential anomalies and normal data along an attribute. By biasing splits toward attributes that yield higher separation, the method can effectively isolate anomalies even when they create dense clusters. This targeted splitting improves detection in datasets with structured anomaly groups while only having a slightly higher time complexity during training. However, this increased complexity during training might create a large bottleneck when working with high-dimensional data.

E. Deep Learning Integration

As iForest methods matured, researchers began exploring how to leverage deep learning to enhance anomaly detection for high-dimensional and nonlinear data. These models aim to enhance detection performance and scalability while addressing the challenge of explainability. This subsection reviews several notable approaches that integrate neural architectures into iForest, including Deep iForest (DIF) [7], Deep Optimal

TABLE V
FEATURE COMPARISON OF CLUSTERING iFOREST VARIANTS

Feature	SCiForest	HiForest	PIF	SaIForest	FlIForest	FForest
Cluster-Aware Splitting	✓	✓	✓		✓	✓
Improved Detection in Dense Regions	✓	✓				
Preference/Embedding Integration			✓			
Tree Selection / Pruning				✓		
Feature Weighting					✓	
Fuzzy Partitioning / Soft Clusters						✓
Interpretability Support	Partial	Limited	Limited	Limited	Partial	Partial
Computational Efficiency	Moderate	Moderate	High	Moderate	High	Moderate
Robustness to Structured Anomalies	High	High	High	High	High	Moderate

iForest (DOIForest) [36], and Attention-Based iForest (ABIForest) [37].

Deep iForest (DIF) [7] was introduced to address iForest's struggle with capturing nonlinear relationships. Instead of relying on axis-aligned splits in raw feature space, DIF uses optimization-free neural networks to generate random feature projections that better separate anomalies from normal data. This process simulates the function of tree rotations or extended splits in other variants, but within a neural framework. To ensure efficiency, DIF uses Computation-Efficient Representation Ensemble (CERE), a method that computes these transformations in parallel using a shared weight matrix, keeping inference costs similar to a standard feedforward neural network. Additionally, DIF proposes Deviation-Enhanced Anomaly Scoring (DEAS), which enhances iForest's traditional path length metric by incorporating a deviation score capturing how far observations deviate from split thresholds, improving sensitivity to subtle outliers.

Deep Optimal iForest (DOIForest) [36] builds on DIF by introducing a genetic optimization process to iteratively refine tree structures. Recognizing that gradient-based learning is unsuitable for non-differentiable tree-based models, DOIForest uses evolutionary algorithms to generate successive generations of improved trees. Each layer of this deep model corresponds to a new generation, similar to a neural network's depth, although the focus remains on improving isolation capacity rather than learning feature representations. Comparative evaluations (Table VI) show that DOIForest frequently outperforms iForest, OptIForest, and DIF on a wide range of benchmarks, demonstrating the value of combining deep optimization with ensemble learning.

Attention-Based Isolation Forest (ABIForest) [37] represents a shift toward explainable deep iForest methods. Rather than modifying tree structures or optimizing them, ABIForest integrates an attention mechanism to compute a weighted average of path lengths across the ensemble. While iForest calculates anomaly scores as a simple average, ABIForest allows the model to assign higher weights to more informative trees based on learned patterns. This is formalized as follows.

$$E[h(x)] = \sum_{t=1}^T \alpha_t \cdot h_t(x), \text{ where } \sum_{t=1}^T \alpha_t = 1 \quad (2)$$

Beyond the scoring mechanism, ABIForest incorporates an attention-based encoder that learns which trees are most relevant for a given input. Although this does not provide

TABLE VI
DOIFOREST RECEIVER OPERATING CHARACTERISTIC (ROC)
PERFORMANCE ADAPTED FROM [36]

Benchmark	AUC			
	<i>iForest</i>	<i>OptiForest</i>	<i>DIF</i>	<i>DOIForest</i>
Arrhythmia	79.7 ± 1.0	79.6 ± 0.8	76.3 ± 1.1	77.8 ± 0.9
Cardio	93.0 ± 0.7	92.8 ± 1.3	93.0 ± 0.5	94.1 ± 1.1
Breastw	95.3 ± 0.1	98.2 ± 0.1	92.1 ± 0.2	98.5 ± 0.1
Pima	66.5 ± 0.7	69.8 ± 0.3	69.6 ± 1.9	71.4 ± 0.8
Backdoor	72.7 ± 2.9	92.7 ± 0.5	92.0 ± 0.5	88.9 ± 1.2
KDDCup99	97.0 ± 0.6	97.4 ± 0.1	88.5 ± 0.7	97.6 ± 0.3
Campaign	70.9 ± 1.0	74.7 ± 0.4	69.3 ± 0.9	74.8 ± 0.4
Creditcard	94.6 ± 0.5	84.8 ± 2.4	94.5 ± 0.7	95.6 ± 0.3
Celeba	69.4 ± 2.4	79.2 ± 1.9	67.9 ± 1.7	80.3 ± 1.0
Mnist	80.2 ± 1.8	85.5 ± 1.0	83.7 ± 1.3	86.0 ± 0.9

Bold indicates the best performing model for a given benchmark, if there is no bolded value present in the row, a non-iForest anomaly detection method performed the best in the original study.

a full explanation of each tree’s decisions, it offers some explainability through a partial view of which regions of the input space are influential.

While DIF and DOIForest prioritize better detection capability through latent representations and structural optimization, ABIForest introduces a novel approach of using attention for interpretability. These methods signal a clear trend toward hybrid architectures that seek to balance the precision of deep models with the practical demands of transparency and deployment efficiency. However, they also introduce challenges in real-time applications where inference speed, model complexity, and interpretability are critical. For fraud detection scenarios that require auditability and low latency, future research is needed to find ways to preserve the benefits of deep integration without sacrificing the strengths of iForest.

F. Explainable iForest

In high-stakes domains like fraud detection, explainability is essential for both regulatory compliance and user trust. Several recent extensions of Isolation Forest explicitly address this need by uncovering which features or data structures contribute most to anomaly detection. These interpretable variants, DIFFI [10], ExIFFI [11], and Decision Predicated Graphs [38], offer tools that help domain experts validate, audit, and understand anomalous outcomes beyond black-box scores.

Depth-Based Feature Importance for iForest (DIFFI) [10] was the first method to build on iForest with the intention of improving explainability. It aims to help analyze and quantify why a data point is labeled as is – a critical requirement in regulated domains like fraud detection. DIFFI computes feature importance by analyzing the distribution of path lengths in the iTrees, determining how much each feature contributes to isolation. These contributions are averaged across trees to generate a ranked list of influential features. In evaluation, DIFFI effectively identified relevant attributes in benchmark datasets, with equal performance to state-of-the-art explainability methods for other anomaly detection forms.

This interpretability aligns with fraud detection’s need for auditability and transparency in decision-making and the results also demonstrated the possibility of real-time deployment. However, DIFFI currently is only deployable on traditional iForest meaning that the robustness of the model is limited by iForest performance.

Extended iForest Feature Importance (ExIFFI) [11] builds upon DIFFI by refining feature importance estimation through extended isolation mechanisms and an adjusted scoring function, making it compatible with EIF. Its goal is to provide the same kind of feature importance infrastructure that DIFFI offers, but adapted to the architecture of EIF. The method recalculates feature contributions based on the random hyperplanes used by EIF ensuring that importance scores remain meaningful even when splits involve multiple features. This adaptation allows analysts to interpret anomaly scores in models that capture more complex separation boundaries. In performance evaluations, ExIFFI maintained the detection strength of EIF while producing stable and relevant importance rankings. For fraud detection, this means a more representative explanation in datasets with intertwined behavioral variables. However, the added complexity of tracking feature contributions across hyperplane splits increases computational overhead, which may limit suitability for real-time fraud detection pipelines.

Decision Predicated Graph iForest (DPG-iForest) [38] differs from DIFFI and ExIFFI by employing a graph structure to improve interpretability. DPG-iForest enhances interpretability by representing iForest’s decision process as a graph structure, mapping how features and their value ranges lead to isolation decisions. This approach aims to give users a visual, step-by-step trace of anomaly detection logic. It constructs a decision graph from the paths taken in iTrees and the nodes represent feature splits and the edges represent decision paths with frequency ratings, enabling users to follow the exact route to anomaly classification. Experiments show that DPG-iForest preserves iForest’s accuracy while significantly improving transparency. In fraud detection, this graphical reasoning could aid investigators in justifying model outcomes to compliance officers and non-technical stakeholders. However, the interpretability benefits come at the cost of higher memory usage and potentially complex graphs, possibly increasing run time.

TABLE VII
FEATURE COMPARISON OF EXPLAINABLE IFOREST VARIANTS

Feature	DIFFI	ExIFFI	DPG-iForest
Supports Extended iForest Splits		✓	
Feature Importance Scoring	✓	✓	
Tree-Path Visualization			✓
Focus on Feature Contribution	✓	✓	
Adaptable to Variant Architectures	Limited	✓	Limited
Anomaly Contextualization	Partial	Partial	✓
Computational Efficiency	High	High	Moderate
Regulatory Compliance Support	✓	✓	✓

The comparison in Table VII illustrates the state of explainable iForest variants. DIFFI and ExIFFI excel at quantifying feature importance, with ExIFFI extending these capabilities to models employing hyperplane splits, making it suitable for EIF architectures. In contrast, Decision Predicated Graph iForest focuses on structural visualization of decision paths, offering anomaly contextualization. Together, these methods address different aspects of the challenge of explanation in fraud detection, although no single approach provides a universal framework for interpretability and real-time deployment.

V. FRAUD DETECTION IN PRACTICE: MISALIGNMENTS & GAPS

In this section, we analyze how the studies of the 24 methods covered above stack up against the domain requirements of fraud detection, and dive into other areas of fraud detection that are under explored. For effective fraud detection, we have defined the requirements to be accuracy, adaptability, scalability, deployability in real-time across multiple devices [40] and interpretability. We define some Key Performance Indicators (KPIs) for each of these requirements:

- **Accuracy:** We define as average $AUC \geq 0.70$; this threshold is based on the average performance of ORCA, a distance-based outlier detector used in [1] as a benchmark for iForest performance.
- **Adaptability:** We define as $\leq 10\%$ performance drop on unseen data, as iForest is designed for unseen data.
- **Online:** We define as capable of retraining and incorporating new data for new trends.
- **Scalability:** We define as keeping the run time during training with large datasets near linear ($< O(n^2)$), like the iForest run time of $O(n \log n)$ [1].
- **Deployability in real-time:** We define as achieving near real-time response with maximum latency ≤ 10 seconds, based on the average time a user will wait with attention on a task [41]. This is measured by comparing run time to iForest, which meets the run time criteria.
- **Interpretability:** We define as having some metric of understanding the decision at each node.

Currently, there is no universal solution among the iForest variants, especially in the domain of fraud detection due to the breadth of the strict requirements. Table VIII outlines the domain requirements that each method meets and contains details of whether the study employed at least one financial benchmark (Creditcard, Campaign, Fraud, etc.). As seen in Table VIII, there are very few methods that satisfy the majority of criteria, and there is no solution that meets all the criteria of fraud detection.

A. Over Emphasis on Financial Fraud

Although there is vast literature on iForest, very few studies investigate the application in fraud-related benchmarks. Among the 24 methods reviewed, only five of them have been explicitly evaluated against fraud-based or financial benchmarks. We hypothesize that this is due to the lack of availability of financial datasets during early research in the

field and the desire for consistent benchmarking. Among the financial benchmarks used in iForest studies, two are related to credit card fraud, and the third looks at retail advertising campaigns; due to the limitation to three financially related datasets, we believe there is a significant gap between the research domain and the real-world deployment of iForest in the fraud domain. We believe that this gap provides an opportunity to assess these variants on the Fraud Dataset Benchmark (FDB) [42].

As we have previously touched on, the financial benchmarks used in iForest studies only represent a fraction of the broader fraud landscape, limiting their scope to payment or credit card fraud. Emerging domains such as e-commerce fraud, insurance fraud, telecom fraud, social media fraud, and cybersecurity scams each present unique operational contexts and challenges in which iForest has not been thoroughly investigated. In the domain of iForest research, some topics of fraud detection have been explored:

- **Insurance fraud** [43]: iForest has been investigated as a post hoc monitoring model for detecting anomalies in medical insurance claims.
- **Cybersecurity** [44]: iForest has been applied to detecting anomalies in web traffic, which serves as a means for identifying malicious activities.
- **Loan Application** [45]: iForest has been applied in a comparative study into anomalies in loan applicant data

Beyond just the research employing iForest, research in fraud detection has explored broader applications:

- **E-commerce tax evasion fraud** [46]: Machine learning models have been employed to detect under-reporting of income, addressing challenges associated with sparse and noisy data.
- **Transactional fraud in food delivery services** [47]: Models have been developed to identify fraudulent behaviors such as false claims of non-delivery.
- **Insurance claims linked to social networks** [48]: Incorporating social networks of claimants, brokers, and garages has been shown to enhance fraud detection beyond claim-level features alone.

These examples highlight that, while iForest research is extensive in certain domains, it remains narrow in scope when considering the diverse and interconnected nature of fraud. Given that fraud rings often operate across multiple domains, expanding anomaly detection research into underexplored areas would not only fill important methodological gaps but would also foster a more comprehensive fraud detection ecosystem. We believe that the first step towards bridging these gaps lies in the creation and utilization of diverse financial and fraud benchmarks that capture the intricacies of fraud networks, allowing for a concrete evaluation of performance, especially in the metric of interpretable artificial intelligence systems.

TABLE VIII
COMPARISON OF iFOREST METHODS REVIEWED

Method	Accuracy	Adaptability	Online	Scalability	Deployability	Interpretability	Financial
iForest [1]	✓	✓		✓	✓		
Rotated iForest [21], [22]	✓	✓		*	*		
Extended iForest [21]	✓	✓		✓	*		
iForestASD [9]	✓	✓	✓	*	*		
IDForest [24]	✓	✓	✓	✓	✓		
Online iForest [25]	✓	✓	✓	✓	✓		
iMondrian Forest [26]	✓	✓	✓	✓	✓		
PCA Variant iForest [6]	✓	✓		*	*		
siForest [27]	✓	✓		*	*		
Generalized iForest [28]	✓	✓		*	*	✓	✓
OptiForest [30]	✓	✓		*			✓
LSHiForest [29]	✓	✓		✓	✓		
Fuzzy Logic based iForest [31], [32]	✓	✓		*	*	✓	
FLiForest [5]	✓	✓		*	*	✓	
SAiForest [4]	✓	✓		✓	✓		
Preference iForest [33]	✓	✓		✓	✓		
Hybrid iForest [34]	✓	✓		*	*		
SCiForest [35]	✓	✓		✓	✓		
Deep iForest [7]	✓	✓			✓		✓
DOIForest [36]	✓	✓			✓		✓
ABIForest [37]	✓	✓		*	*	✓	✓ ^a
DIFFI [10]	✓	✓		*	✓	✓	
ExIFFI [11]	✓	✓		✓	✓	✓	
DPG-iForest [38]	✓	✓		*	*	✓	

* Indicates that a training time complexity and/or run time analysis was not performed in the study.

^aFinancial fraud detection investigated in [39]

VI. CHALLENGES, OPEN PROBLEMS & THE FUTURE OF iFOREST IN FRAUD DETECTION

The challenge of anomaly detection research is to find a solution that checks all the boxes of the use case. In the case of fraud detection, we defined six KPIs and among the current state-of-the-art iForest solutions none satisfy all six KPIs and only six satisfy four or more KPIs. Although there is no universal solution today, the current literature reviewed has shown great diversity in capability and steps toward the domain requirements. Based on the aforementioned requirements of banking fraud detection, especially the current limitations of handling streaming data and interpretability, we conclude that iForest in and of itself is not an adequate approach for banking fraud detection. Our current research focuses on addressing these shortcomings for a more powerful and integrated iForest approach. We believe that therein lies an opportunity to develop an anomaly detection method that utilizes the isolation principle of iForest but employs a data structure more aligned with interpretability and adaptability.

Building on these observations, we argue that the future of fraud detection may lie in hybrid anomaly detection methods that combine the robustness of iForest variants and more explainable data structures with the emerging advantages of quantum anomaly detection techniques used in banking. Quantum methods, while still in their infancy, have shown potential to accelerate subroutines such as sampling, search (via Grover's algorithm), and probability estimation (via quantum Fourier

transforms), which are central to iForest variants. Integrating these quantum capabilities with iForest-inspired models could enable algorithms to update adaptively, maintain robustness under adversarial conditions, and operate in real-time while preserving a more robust model. Such a hybrid paradigm would address the key performance indicators we identified and represent a promising path forward for practical fraud detection deployments. Recent work on quantum-enhanced anomaly detection has further highlighted this potential, suggesting that future research should explore hybrid approaches where classical isolation-based structures interface with quantum subroutines for scalability and speed [49].

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